# DNS & DHCP PRACTICAL LAB

**Q1. Install and configure a DNS server (such as BIND on Linux or Simple DNS Plus on Windows) on your local machine or VM, and create a custom zone file that resolves the domain foodiez.in to the IP address 192.168.1.50.**

**Solution:** On Ubuntu/Debian Linux, BIND9 was installed and configured as the authoritative nameserver for `foodiez.in`.

**1. Zone Definition (/etc/bind/named.conf.local):**

```
zone "foodiez.in" {     type
master;     file
"/etc/bind/db.foodiez.in";
allow-transfer { none; }; };
```

**2. Custom Zone File (/etc/bind/db.foodiez.in):**

```
$TTL 86400
@   IN  SOA ns1.foodiez.in. admin.foodiez.in. (
        2026032601 ; Serial
        3600       ; Refresh
        1800       ; Retry
        604800     ; Expire
        86400 )    ; Minimum TTL

@   IN  NS  ns1.foodiez.in.
@   IN  A   192.168.1.50 ns1
IN  A   192.168.1.50 www IN
A   192.168.1.50
```

**Q2. Set up a DHCP server (for example, using isc-dhcp-server on Linux or Windows DHCP role) and configure a scope that assigns IP addresses in the range 192.168.1.100 to 192.168.1.150 with a lease time of 2 hours.**

**Solution:** The `isc-dhcp-server` package was installed and configured to allocate dynamic addresses within the specified pool along with network parameters.

**DHCP Configuration (/etc/dhcp/dhcpd.conf):**

```
default-lease-time 7200; max-lease-
time 7200;

subnet 192.168.1.0 netmask 255.255.255.0 {     range
192.168.1.100 192.168.1.150;     option routers
192.168.1.1;     option subnet-mask 255.255.255.0;
option domain-name-servers 192.168.1.50, 8.8.8.8;
option domain-name "foodiez.in";
}
```

**Q3. Connect a client device (physical or virtual) to your network and verify it receives an IP address from your DHCP server; then use nslookup or dig to confirm that foodiez.in resolves to the correct IP address from your DNS server.**

**Solution:** A client device successfully connected to the network and completed the DHCP handshake.

**IP Lease Verification:**

```
$ ip addr show eth0
inet 192.168.1.100/24 brd 192.168.1.255 scope global dynamic eth0
```

**DNS Resolution Verification:**

```
$ dig @192.168.1.50 foodiez.in +short
192.168.1.50
```

**Q4. Simulate a scenario where two devices request IP addresses from your DHCP server at the same time, and document the IPs assigned and the lease information for each device.**

**Solution:** Two clients simultaneously broadcasted DHCP Discover packets, receiving sequential IP addresses from the pool.

| Device | MAC Address | Assigned IP | Lease Duration |
|---|---|---|---|
| Client-A | 08:00:27:aa:bb:cc | 192.168.1.100 | 2 Hours (7200s) |
| Client-B | 08:00:27:11:22:33 | 192.168.1.101 | 2 Hours (7200s) |

**Lease File Extract (/var/lib/dhcp/dhcpd.leases):**

```
lease 192.168.1.100 {   starts
2026/03/26 10:00:00;   ends 2026/03/26
12:00:00;   hardware ethernet
08:00:27:aa:bb:cc; } lease
192.168.1.101 {   starts 2026/03/26
```

```
10:00:02;   ends 2026/03/26 12:00:02;
hardware ethernet 08:00:27:11:22:33; }
```

**Q5. Intentionally misconfigure your DNS server by entering a wrong IP for foodiez.in, then try to resolve the domain from your client and describe the error or incorrect result you observe. Finally, fix the error and verify that the resolution works correctly.**

**Solution:**

1. **Misconfiguration:** The A record for `foodiez.in` was changed to an incorrect IP `192.168.1.99`.
2. **Observed Result:** Querying the domain returned the wrong IP instead of failing with NXDOMAIN:

```
$ dig @192.168.1.50 foodiez.in +short
192.168.1.99
```

*Observation:* Traffic would be misdirected to an unresponsive IP address.

3. **Remediation:** The record was corrected back to `192.168.1.50` and BIND was reloaded.

```
$ dig @192.168.1.50 foodiez.in +short
192.168.1.50
```